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FLOATING SUPPORT GRID SYSTEM FOR HORIZONTAL AMMONIA CONVERTER BASKET CATALYST BED

Abstract

A floating support grid system including a plurality of support beams and a plurality of profile wire screen panels wherein the support beams are connected to the profile wire screen panels via one or more first floating connections, wherein the support beams are configured to connect to a horizontal ammonia converter basket via one or more second floating connections, and wherein at least some the profile wire screen panels are configured to connect to a horizontal ammonia converter basket via one or more third floating connections, wherein each of the first, second, and third floating connections independently defines one or more gaps to accommodate movement caused by thermal expansion.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION

[0001] This application claims priority to India Application No. 202411016961, filed Mar. 8, 2024, which is incorporated by reference herein.

FIELD OF THE INVENTION

[0002] The present invention relates to a floating support grid system. In examples, the disclosure herein relates to a floating support grid system for a horizontal ammonia converter basket catalyst bed.

DISCUSSION OF THE RELATED ART

[0003] A horizontal ammonia converter basket is a key component in the process of synthesizing ammonia through the Haber-Bosch process. It consists of a cylindrical vessel typically made of stainless steel, horizontally oriented, and filled with catalyst material. The catalyst material, often based on iron with promoters like alumina and potassium oxide, facilitates the conversion of nitrogen and hydrogen into ammonia.

[0004] The reactants, nitrogen, and hydrogen, are passed through the catalyst bed under high pressure and temperature, typically ranging from 150 to 300 atmospheres and 350° C. to 500° C., respectively. The catalyst promotes the desired chemical reaction, producing an ammonia-rich product stream.

[0005] Horizontal converter baskets are used in large-scale industrial ammonia production plants due to their efficiency and scalability. They allow for continuous operation and high throughput of ammonia synthesis, crucial for meeting the demands of various industries reliant on ammonia as a feedstock for fertilizers, chemicals, and other applications. In a horizontal ammonia converter basket, catalyst supports play a crucial role in facilitating the ammonia synthesis reaction. Their primary function is to provide a stable structure for the catalyst bed, ensuring uniform distribution of the catalyst material throughout the basket. Additionally, these supports help optimize the surface area available for the catalytic reaction, enhancing the efficiency of ammonia production. The design of the catalyst supports in a horizontal converter basket is essential for maintaining optimal reaction conditions and maximizing the conversion of reactants into ammonia.

[0006] The catalyst in a horizontal converter basket is typically supported by a support grid system. In known systems, the catalyst support grid system has a profile wire screen with integral beams that act as a single unit. In case of mal-operational or severe plant upsets, the support grid system is subjected to rapid changes in temperature (beyond permissible range). In such a scenario, the beams are required to expand and move within the catalyst bed. Such expansions and movements may be resisted by the catalyst leading to resistance and excessive stresses in the support grid system consequently leading to failure of the support grid system. Also, catalyst containment structure is often provided using cover plates near the wall, partition plates, and/or end plates. These cover plates are in direct contact with the profile wires. During thermal expansion of the support grid system, the profile wires may not slide under the cover plates and may get stuck, which is undesirable.

[0007] When a plant experiences frequent upsets, the current support grid system with profile wire screens welded to beam is subjected to excessive thermal loads. These high thermal loads can lead to failure of the profile wire screen panel or the weld-joints between the beam and the profile wire

screen panel.
[0008] Thus, an improved catalyst support grid system can be useful in providing a more stable structure.

SUMMARY

[0009] Examples of floating support grid system can substantially obviate one or more of the problems due to limitations and disadvantages of the related art or at least to provide the public with a useful alternative.
[0010] Additional features and advantages of the examples as described herein will be set forth in the description which follows, and in part will be apparent from the description, or may be learned by practice of the examples described herein. The objectives and other advantages will be realized and attained by the structure particularly pointed out in the written description and claims hereof as well as the appended drawings.
[0011] To achieve these and other advantages and in accordance with examples described herein, as embodied and broadly described, is a floating support grid system and a horizontal ammonia converter including a floating support grid system.
[0012] In examples, disclosed herein is a floating support grid system that may include a plurality of support beams; a plurality of profile wire screen panels; a plurality of first connections each connecting one support beam of the plurality of support beams to one profile wire screen panel of the plurality of profile wire screen panels, wherein, at least one first connection of the plurality of first connections may include a first floating connection configured to define a first gap between the one support beam of the plurality of support beams and the one profile wire screen panel of the plurality of wire screen panels the at least one first connection connects; and a plurality of second floating connections each configured to connect a support beam of the plurality of support beams to a horizontal ammonia converter basket.
[0013] In examples, the first gap may be configured to accommodate a thermal expansion movement of a profile wire screen panel, a support beam, or both.
[0014] In examples, each first floating connection may connect one side of the one profile wire screen panel to one side of the one support beam.
[0015] In examples, at least one profile wire screen panel of the plurality of profile wire screen panels may be connected to a first support beam of the plurality of support beams at a first lateral side of the at least one profile wire screen panel and may be connected to a second support beam of the plurality of support beams at a second lateral side, opposite the first lateral side, of the at least one profile wire screen panel.
[0016] In examples, one first floating connection may be provided to connect the first support beam with the at least one profile wire screen panel at the first lateral side, and another first floating connection may be provided to connect the second support beam with the at least one profile wire screen panel at the second lateral side.
[0017] In examples, every first connection may include a first floating connection.
[0018] In examples, each profile wire screen panel of the plurality of wire screen panels may include a wire grid may include one or more wires; at least one support bar provided below the wire grid; and a built-in plate over a peripheral area of the wire grid.
[0019] In examples, the first floating connection further including a first cover plate configured to overlap at least a portion of the built-in plate over the peripheral area of the wire grid of the one profile wire screen panel connected to the one support beam, wherein the first cover plate may extend from the one support beam toward the one profile wire screen panel connected to the one support beam.
[0020] In examples, the floating support grid system may include the one or more cleats extending from a vertical web portion of the one support beam connected to the one profile wire screen panel to a top surface of the first cover plate.
[0021] In examples, the first floating connection further may include a first anti-lift system,

wherein the first anti-lift system may include an anti-lift bar fixed connected to a bottom surface the one profile wire screen panel, the anti-lift bar configured to extend below a bottom surface of a base portion of the one support beam connected to the one profile wire screen panel. In examples, one or more stoppers extending from the base portion of the one support beam may be connected to the one profile wire screen panel, the one or more stoppers may be arranged to be on one or more lateral sides of the first anti-lift system to prevent lateral movement of the one profile wire screen panel relative to the one support beam connected thereto.

[0022] In examples, the floating support grid system may include one or more third floating connections configured to connect at least one profile wire screen panel of the plurality of wire screen panels to the horizontal ammonia converter basket, wherein each third floating connection may be configured to define a third gap between the at least one profile wire screen panel and a third or fourth portion of the horizontal ammonia converter basket it may be configured to connect when the at least one profile wire screen panel may be connected to the horizontal ammonia converter basket. In examples, the third gap may be configured to accommodate a thermal expansion movement of the at least one profile wire screen, a support beam of the plurality of support beams, or both. In examples, every third floating connection may include a second cover plate arranged to extend over the third gap defined by third floating connection; and a second anti-lift system.

[0023] In examples, at least one second floating connection of the plurality of second floating connections may be configured to define one or more second gaps between the support beam of the plurality of support beams and a second one or more portions of the horizontal ammonia converter basket when the at least one second floating connection connects a lateral end of the support beam of the plurality of support beams to the horizontal ammonia converter basket.

[0024] In examples, each of the one or more second gaps may be configured to independently accommodate a thermal expansion movement of the support beam of the plurality of support beams, a profile wire screen panel of the plurality of profile wire screen panels, or both.

[0025] In examples, the support beam of the plurality of support beam may include a first lateral end and a second lateral end, opposite the first lateral end, and wherein one second floating connection may be arranged to connect the first lateral end to the horizontal ammonia converter basket and another second floating connection may be arranged to connect the second lateral end to the horizontal ammonia converter basket.

[0026] In examples, every support beam of the plurality of support beams may include a respective first lateral end with a respective second floating connection and a second lateral end, opposite the first lateral end, with a respective second floating connection, wherein every first lateral end and every second lateral end are connected to the horizontal ammonia converter basket.

[0027] In examples, the second floating connection may be configured to define a panel lateral side gap between a profile wire screen panel connected to the support beam of the plurality of support beams and one of the second portions of the horizontal ammonia converter basket. In examples, the panel lateral side gap may be below a lateral ledge fixed connected to the horizontal ammonia converter basket.

[0028] In examples, the second floating connection may include one or more abutting plates each fixed connected to the support beam of the plurality of support beams and one or more lateral ledge plate extending form a lateral ledge fixed connected to the horizontal ammonia converter basket, wherein each abutting plate may be aligned with a lateral ledge plate and configured to define a respective beam lateral side gap when the support beam of the plurality of support beams may be connected to the horizontal ammonia converter basket. In examples, one or more second cover plates may be provided, each second cover plate configured to cover the respective beam lateral side gap. In examples, the respective second cover plate may be configured to overlap with at least a portion of the respective lateral ledge plate and to slide over the respective lateral ledge plate.

[0029] In examples, the second floating connection provided at a lateral end of a support beam may include two abutting plates and two lateral ledge plates, wherein the two abutting plates and the two lateral ledge plates are arranged to define two separate beam lateral side gaps. In examples, an integral cover plate may be arranged to cover both separate beam lateral side gaps and extending over both lateral ledge plates.

[0030] In examples, the second floating connection may be configured to define a first beam web lateral end gap between a vertical web portion of the support beam and a lateral ledge fixed connected to the horizontal ammonia converter basket. In examples, a third cover plate may cover the first beam web lateral gap, wherein the third cover plate may be configured to slide over the lateral ledge.

[0031] In examples, the second floating connection may be configured to define a second beam web lateral gap between a vertical web portion of the support beam and one of the second portions of the horizontal ammonia converter basket, wherein the second beam web lateral gap may be below a lateral ledge fixed connected to the horizontal ammonia converter basket.

[0032] In examples, the second floating connection may be configured to define a support beam base lateral end gap between a base portion of the support beam and one of the second portions of the horizontal ammonia converter basket.

[0033] In examples, the second floating connection may be configured to define an anti-lift bar gap between an anti-lift bar connected to a base portion of the support beam and a mating structure connected to the horizontal ammonia converter basket.

[0034] In examples, provided is a floating support grid system including a plurality of support beams, each support beam including a first lateral end and a second lateral end, opposite the first lateral end; a plurality of profile wire screen panels each connected via a respective first floating connection to at least one support beam of the plurality of support beams, each profile wire screen panel including: a first lateral side edge and a second lateral side edge; a first longitudinal side edge and a second longitudinal side edge; a grid including one or more wires; and a built-in plate over a peripheral area of the grid; one second floating connection configured to connect a first lateral side edge of a first profile wire screen panel to a horizontal ammonia converter basket; another second floating connection configured to connect a second lateral side edge of a second profile wire screen panel to the horizontal ammonia converter basket; and a third floating connection provided at each first lateral end of each support beam and at each second lateral end of each support beam, wherein the third floating connection may be configured to connect each respective first lateral end and second lateral end to the horizontal ammonia converter basket, wherein, each first floating connection may be configured to define a first gap between each profile wire screen panel and respective support beam to which it may be connected, wherein, each second floating connection may be configured to define a second gap between the respective first lateral side edge or second lateral side edge and a first or second portion of the horizontal ammonia converter basket, wherein each third floating connection may be configured to define respectively one or more third gaps between the first lateral end of each support beam and one or more third portions of the horizontal ammonia converter basket, between the second lateral end of each support beam and one of the third portions of the horizontal ammonia converter basket, between the first longitudinal side edge of each profile wire screen panel and one of the third portions of the horizontal ammonia converter basket, and between the second longitudinal side edge of each profile wire screen panel and one of the third portions of the horizontal ammonia converter basket, and wherein the first gap, the second gap, and the one or more third gaps are configured to accommodate a thermal expansion movement of one or more support beams, profile wire screen panels, or both.

[0035] In examples, provided is a horizontal ammonia converter including a horizontal ammonia converter basket; and a floating support grid system as described herein.

[0036] In examples, provided is a floating support grid system and a horizontal ammonia converter including a floating support grid having any one or more of the above features in any combination as more fully described herein.

[0037] It is to be understood that both the foregoing general description and the following detailed description are exemplary and explanatory and are intended to provide further explanation of the examples as claimed.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0038] The accompanying drawings, which are included to provide a further understanding of the examples and are incorporated in and constitute a part of this specification, illustrate embodiments of the invention and together with the description serve to explain the principles of the invention.

[0039] In the drawings:

[0040] FIG. **1**A illustrates a diagram of an example of a horizontal ammonia converter basket in which a floating support grid system as described may be employed.

[0041] FIG. **1**B illustrates a prior art catalyst bed design.

[0042] FIG. **1**C illustrates a prior art design for a profile screen wire panel.

[0043] FIG. **1**D illustrates a prior art connection between a profile screen wire panel and two support beams.

[0044] FIGS. **2**A-**2**C illustrate diagrams of examples of a floating support grid system as described herein.

[0045] FIG. **2**D illustrates a diagram of an example of a profile screen wire panel that may be used in the floating support grid system described herein.

[0046] FIGS. **3**A-**3**C illustrate front, side, and plan view diagrams of an example floating connection between one or more profile screen wire panel and a support beam that may be implemented in the floating support grid system as described herein.

[0047] FIG. **4** illustrates a diagram of an example of a floating connection between a lateral side of a profile screen wire panel and a wall or other structure of a horizontal ammonia converter basked as may be implemented in a floating support grid system described herein.

[0048] FIGS. **5**A-**5**C illustrate diagrams of an example floating connection between a lateral end of a floating support grid system and a wall or other structure of a horizontal ammonia converter basket.

DETAILED DESCRIPTION OF THE ILLUSTRATED EMBODIMENTS

[0049] Disclosed is a floating support grid system. In examples, the system as described provides a floating design for support grid system arrangement in a horizontal ammonia converter basket. In examples, disclosed is a floating support grid system configured to allow for one or more profile support grid wire screen panels to freely expand and contract even with rapid changes in temperature while still maintaining a catalyst containment arrangement. In examples, one or more support beams configured to support one or more profile wire screen panels may be configured to expand with temperature with little to no effect on the profile wire screen panels.

[0050] In examples, the floating support grid system may include one or more floating connections that define and/or include one or more gaps to accommodate for thermal expansion movements in the longitudinal direction and/or the lateral direction of one or more components of the floating support grid system.

[0051] In examples, the floating support grid system may include one or more support beams joined to the horizontal ammonia converter basket and/or to one or more profile wire screen panels through methods other than welding. In examples, the one or more beams may be connected to one or more profile wire screen panels via one or more floating connections. In examples, the one or more beams may be connected to the horizontal ammonia converter basket via one or more floating connections. In examples, the floating support grid system may include every connection to every support beam to include a floating connection. In examples, the floating support grid system may include some but not all connections to every support beam to define and/or include a floating connection. In examples, one or more floating connections to a support beam may define and/or include one or more gaps to accommodate for thermal expansion movement of one or more support beams, one or more profile wire screen panels, or a combination of both. In examples, every floating connection to a support beam may include one or more gaps to accommodate for thermal expansion movement of one or more support beams, one or more profile wire screen panels, or a combination of both.

[0052] In examples, the floating support grid system may include one or more profile wire screen panels joined to the horizontal ammonia converter basket and/or to one or more support beams through methods other than welding. In examples, the one or more profile wire screen panels may be connected to one or more support beams via one or more floating connections. In examples, the one or more profile wire screen panels may be connected to the horizontal ammonia converter basket via one or more floating connections. In examples, the floating support grid system may include every connection to every profile wire screen panel to include a floating connection. In examples, the floating support grid system may include some but not all connections to every profile wire screen panel to include a floating connection. In examples, one or more floating connections to a profile wire screen panel may define and/or include one or more gaps to accommodate for thermal expansion movement of one or more support beams, one or more profile wire screen panels, or a combination of both. In examples, every floating connection to a profile wire screen panel may define and/or include one or more gaps to accommodate for thermal expansion movement of one or more support beams, one or more profile wire screen panels, or a combination of both.

[0053] As used herein, the term “floating connection” refers to a connection that allows controlled movement or flexibility between the two connected components. Unlike rigid connections, which are fixed in place, a floating connection permits a certain degree of motion, such as thermal expansion, while maintaining secure attachment. A floating connection may include one or more components arranged to form a fit connection or sliding or flexible components that permit controlled movement. As used herein, floating connection excludes weld-joints or fixed connections such as those made by fasteners like screws, bolts, pins, or other like structures, and/or by welding, adhesive or like means.

[0054] In examples, the floating support grid system as described may define, include, and/or maintain one or more separation gaps for the support beams and/or for the profile wire screen panels to allow for differential expansion between the beams and the profile wire screen panels. In examples, one or more gaps between the beams and the profile wire screen panels may be provided to allow for differential expansion movement between the beam and the profile wire screen panels. In examples, one or more gaps the beams and the walls or other portion of the horizontal ammonia converter basket may be provided to allow for expansion movement of the beams. In examples, one or more gaps may be provided between a profile wire screen panel and the walls or other portion of the horizontal ammonia converter basket to allow for expansion movement of the profile wire screen panel.

[0055] In examples, the gaps may be covered by cover plates to prevent catalyst or other debris from entering the gaps and causing migration of catalyst. In examples, individual profile wire screen panels may be protected on the sides with built-in plates to avoid or minimize rubbing and thus shear stress between cover plate and the wires of the profile wire screen panel. In examples, the floating support grid system as described may include one or more anti-lift mechanisms. In examples, one or more anti-lift mechanisms may be provided to prevent lifting of one or more profile wire screen panels. In examples, one or more anti-lift mechanisms may be provided to prevent lifting of one or more support beams.

[0056] In examples, the horizontal ammonia converter basket equipped with a floating support grid system as described may be able to withstand higher rates of temperature changes typically seen in uncontrolled upset conditions. This may help safeguard the equipment and prevent failures and/or loss of catalyst containment functionality. For purposes of this disclosure "catalyst containment" as referring to the functionality of the floating support grid system refers to the catalyst substantially remaining on the floating support grid system as opposed to falling through in an unintended manner.

[0057] Reference will now be made in detail to one or more examples, which are illustrated in the accompanying drawings.

[0058] Unless defined otherwise, all technical and scientific terms used herein have the same meaning as is commonly understood by one of skill in the art to which the inventions belong. All patents, patent applications, published applications and publications, websites and other published materials referred to throughout the entire disclosure herein, unless noted otherwise, are incorporated by reference in their entirety. Where there is a plurality of definitions for terms herein, those in this section prevail. Where reference is made to a URL or other such identifier or address, it is understood that such identifiers can change and information on the internet can come and go, but equivalent information can be found by searching the internet. Reference thereto evidences the availability and public dissemination of such information.

[0059] As used herein, the singular forms "a", "an" and "the" include plural referents unless the context clearly dictates otherwise.

[0060] The terms first, second, third, etc. as used herein can describe various elements, components, regions, layers and/or sections, these elements, components, regions, layers and/or sections should not be limited by these terms. These terms may be only used to distinguish one element, component, region, layer or section from another region, layer, or section. Terms such as "first", "second", and other numerical terms when used herein do not imply a sequence or order unless clearly indicated by the context. Thus, a first element, component, region, layer, or section discussed below could be termed a second element, component, region, layer, or section without departing from the teachings of the example embodiments.

[0061] As used herein, ranges and quantities can be expressed as "about" a particular value or range. "About" also includes the exact amount. Hence "about 5 percent" means about 5 percent in addition to 5 percent. The term "about" means within typical experimental error for the application or purpose intended.

[0062] As used herein, "and/or" includes any and all combinations of one or more of the associated listed items.

[0063] As used herein, a "combination" refers to any association between two items or among more than two items. The association can be spatial or refer to the use of the two or more items for a common purpose.

[0064] As used herein, "comprising" and "comprises" are to be interpreted to mean "including but not limited to" and "includes but not limited to", respectively.

[0065] As used herein, "optional" or "optionally" means that the subsequently described event or circumstance does or does not occur, and that the description includes instances where the event or circumstance occurs and instances where it does not. For example, an optional component in a system means that the component may be present or may not be present in the system.

[0066] As used herein, "substantially" means "being largely but not wholly that which is specified."

[0067] FIG. **1**A provides a diagram of an example of a horizontal ammonia converter basket **104** in which the floating support grid system as described herein may be employed. The illustration is only an example. The floating support grid system as described herein may be employed in other horizontal ammonia converter basket designs. Non-limiting examples of horizontal ammonia converter basket designs in which the floating support grid system as described herein may be employed are described in U.S. Pat. Nos. 4,696,799, 7,867,465, and 6,132,687, all of which are incorporated herein by reference in their entirety.

[0068] In examples, horizontal ammonia converter **100** may generally include a vessel **102** and a horizontal ammonia converter basket **104** inside vessel **102**. In examples, a horizontal ammonia converter basket **104** may include a cylindrical inner portion **106** disposed within vessel **102**, a first end closure **108** and a second end closure **110**. In examples, first end closure **108** and second end closure **110** may be attached to ends of vessel **102** and form first gas plenum **112**, and second gas plenum **114**, respectively. A gas inlet **116**, may be provided to feed relatively cool ammonia synthesis gas into the second gas plenum **114**, a shell annulus **118**, and then to the first gas plenum **112**.

[0069] One or more catalyst beds **120** (e.g. **120***a*, **120***b*, **120***c*, and **120***d*) may be provided inside a horizontal ammonia converter basket **104** vessel. Each catalyst bed **120** may include a top inlet portion **122** and a bottom outlet portion **124**, the catalyst particles being supported by grids and screens not shown. In examples, the catalyst beds **120** may be arranged for downward flow of gas in a path substantially normal to the axis of vessel **102** of the horizontal ammonia converter basket **104**.

[0070] In examples, the horizontal ammonia converter basket **104** may further include one or more tubular heat exchangers **126** (e.g., **126***a* and **126***b*) having respective tube inlet portions **128** (e.g., **128***a* and **128***b*) and tube outlet portions **130** (e.g., **130***a* and **130***b*) located between catalyst beds **120**. In examples, a first tubular heat exchanger **126***a* may include a vertical U-tubes to provide high internal tube gas velocity and, therefore, high heat transfer rates which thereby minimize the size of this exchanger.

[0071] In examples, the horizontal ammonia converter basket **104** may include a flow path through the catalyst beds **120** and first and second exchangers provided by longitudinal conduits within the cylindrical inner portion **106** above and below the catalyst beds **120**.

[0072] As illustrated, gas may be introduced to flow across tube interior of the tubular heat exchangers **126** and then flows upwardly through a channel and into top inlet portion **122** of a first catalyst bed **122***a*. The gas then may be channeled as illustrated to flow to one or more subsequent catalysts beds **122** (e.g., **122***b*, **122***c*, and **122***d*) in a similar manner by flowing into the respective top inlet portion **122** and out the bottom outlet portion **124** of each catalyst bed **122**. Ultimately, the gas may be output from the horizontal ammonia converter basket **104** via an outlet pipe **132**.

[0073] FIGS. **1**B-**1**D illustrate prior art arrangements of support grid or screen for catalyst beds **120**. FIG. **1**B is a diagram of a perspective view. As shown, a catalyst bed **120** may include one or more profile wire screen panels **134** and one or more support beams **136**. In known arrangements, the profile wire screen panels **134** are welded to the support beams **136**. As shown, the weld-joint may be formed along each abutting edge **148** of profile wire screen panels **134**. FIG. **1**D illustrates a side view of a single profile wire screen panel **134** welded to two abutting supporting beams **136** (e.g. **136***a* and **136***b*).

[0074] As illustrated in FIG. **1**C, a profile wire screen panel **134** may include a grid of V wires **142**. In examples, a profile wire screen panel **134** may generally include one or more panel support bars **138** located under and configured to support the grid of wires **142**. As shown, a profile wire screen panel **134** may include an edging bar **140** at each side (e.g. **140***a*, **140***b*, **140***c*, and **140***d*). In examples, the wires **142** are at the top inlet portion **122** of the catalyst bed and are configured to hold the catalyst while allowing the gas to pass through.

[0075] As shown in FIG. **1**D, in known arrangements, the profile wire screen panels **134** rest on at least a portion of a base **144** (e.g. **144***a* and **144***b*) of the support beams **136***a* and **136***b* respectively and are connected to the support beams **136** along each abutting edge **148** by one or more welded joints **150** connecting the edging bars **140** to the support beams **136**, and/or by one or more welded joints **152** connecting at least one panel support bar **138** to the base **144** of support beams **136**.

[0076] The welded joints between the profile wire screen panels **134** and the support beams **136** lead to an overall integral structure.

[0077] As shown in FIG. **1**B, the integral structure including the profile wire screen panels **134** and the support beams **136** is generally fixed at **146** to the cylindrical inner portion **106** of the horizontal ammonia converter basket **104** by only a single support beam, and particularly by the central support beam **136′**.

[0078] With this arrangement, the catalyst bed **120** can risk structural failure or break of catalyst containment. This is because during operation the heat in vessel **102** may cause the profile wire screen panels and/or one or the beams to expand. This thermal expansion gives rise to a motion in the longitudinal direction X, traverse Y direction, or both of the catalyst bed **120**. Being an integral structure, this expansion motion of a profile wire screen panel **134** can induce sliding motion of a support beam **136** along the longitudinal X direction on the whole catalyst bed **120**. In case of mal-operational or sever plan upsets, the support grid system may be subjected to rapid changes in temperature (beyond permissible range). In such a scenario, the support beams are required to expand and move within the catalyst bed. Such expansions and movements may be resisted by the catalyst leading to resistance and excessive stresses in the support grid system consequently leading to failure of the support grid system, which can lead to structural failure or break of catalyst containment.

[0079] To resolve this issue in the prior art systems, the present disclosure provides a floating support grid system in which the profile wire screen panels are not welded to the support beams. In examples, in the floating support grid system as described, all the support beams may be connected to the horizontal ammonia converter basket.

[0080] FIGS. **2**A-**2**C illustrate diagrams of an example of a floating support grid system as described herein in which the profile wire screen panels are not welded to the support beams.

[0081] FIG. **2**A illustrates a plan view of an example of a catalyst bed support implemented by way of a floating support grid system **200** as described herein. In examples, the floating support grid system **200** may include a plurality of profile wire screen panels **202**. In examples, the floating support grid system **200** may include one or more profile wire screen panels **202** (e.g., **202***a*, **202***b*, **202***c*, **202***d*, **202***c*, and **202***f*). In examples, the floating support grid system **200** may include a plurality of support beams **204**. In examples, the floating support grid system **200** may include one or more support beams **204** (e.g., **204***a*, **204***b*, **204***c*, **204***d*, and **204***c*). In examples, a floating support grid system **200** may be implemented to include any number of profile wire screen panels **202** and/or support beams **204**. In examples, the number of profile wire screen panels **202** in a floating support grid system **200** may depend on the overall size of the floating support grid system **200**, the size of the profile wire screen panels **202**, the number and/or size of support beams **204**, and/or the size of the space in which the floating support grid system **200** is to be installed. In examples, the number of support beams **204** in a floating support grid system **200** may vary depending on their size, the number and/or size of the profile wire screen panels **202**, the size and structural strength of the support beams **204**, the overall size of the a floating support grid system **200**, and/or the size of the space in which the a floating support grid system **200** is to be installed. For purposes of illustration, as an example, FIG. **2**A illustrates a floating support grid system **200** with six profile wire screen panels **202** and five support beams **204**.

[0082] In examples, each profile wire screen panel **202** may be supported by one or more support beams **204**. In examples, the floating support grid system **200** may include one or more profile wire screen panels **202** supported by at least two support beams **204**. For example, as shown in FIG. **2**A, profile wire screen panels **202***b*, **202***c*, **202***d*, and **202***e* are each supported by two abutting support beams **204**. As shown in FIG. **2**B, in examples, the floating support grid system **200** may include one or more profile wire screen panels **202** supported by combination of a support beam **204** and one or more portions **212**, e.g. **212***a* and **212***b*, of a horizontal ammonia converter basket, e.g., a wall or other internal structure, of horizontal ammonia converter basket. For example, such profile wire screen panels **202** may be those located at a longitudinal (i.e. in the longitudinal X direction) end portion **210**, e.g. **210***a* and **210***b*, of the floating support grid system **200**. As illustrated in FIG. **2**B, these may include profile wire screen panels **202***a* and **202***f*, wherein each is respectively supported by support beam **204***a* and **204***c* on a respective inner edge side, e.g. **206***a* and **206***f*, while each having a respective outer edge side, e.g. **208***a* and **208***f*, designated to be connected to respective portions **212***a* and **212***b* of horizontal ammonia converter basket such as a wall or inner structure thereof. In examples, at longitudinal end portions **210***a* and **210***b* a floating support grid system **200** may include one or more third floating connections, referred to herein as longitudinal end connections **216** (e.g., **216***a* and **216***b*) configured to directly connect profile wire screen panels **202** to the horizontal ammonia converter basket system. In examples, a longitudinal end connection **216** may include floating connection. In examples, every longitudinal end connection **216** of floating support grid system **200** may include a floating connection. In examples, a longitudinal end connection **216** may include a floating connection that is also an anti-lift connection. In examples, every longitudinal end connection **216** of floating support grid system **200** may include a floating connection that is also an anti-lift connection.

[0083] In examples, one or more of support beams **204** of floating support grid system **200** may be configured to be connected to the horizontal ammonia converter basket. In examples, each and every support beam **204** may be configured to be connected to the horizontal ammonia converter basket. This may be in contrast to the prior art system in which only a central support beam is fixed to the horizontal ammonia converter basket system. In examples, one or more connections between each support beam **204** and the horizontal ammonia convert basket may include a floating connection.

[0084] In examples, a support beam **204** may be connected to a horizontal ammonia converter basket at one or both lateral (i.e. width direction Y) end portions **218** (e.g., **218***a*, **218***b*, **218***c*, and **218***d*) and **220** (e.g., **220***a*, **220***b*, **220***c*, and **220***d*). For example, support beam **204***c*, may be connected to a horizontal ammonia converter basket at lateral end portions **218***c* and **220***c*. In examples, each connection between a support beam **204** and the horizontal ammonia converter basket system at lateral end portions **218** and **220** may be a floating connection. In examples, as described herein, every support beam **204** of floating support grid system **200** may be connected at both respective lateral end portions **218** and **220** to the horizonal ammonia converter basket. In examples, every connection at each lateral end portion **218** and **220** of every support beam **204** may include a floating connection.

[0085] In examples, none of the profile wire screen panels **202** are welded to any of the support beams **204**. In examples, the floating support grid system **200** may include one or more first connections, or a plurality of first connections each connecting a support beam **204** to a profile wire screen panel **202**. In examples, each first connection, referred to herein as beam-panel connection **214** (e.g., **214***a*, **214***b*, **214***c*, **214***d*, **214***c*, **214***f*, **214***g*, **214***h*, **214***i*, and **214***j*) between a profile wire screen panel **202** and a support beam **204** may include a floating connection. In examples, a beam-panel connection **214** may include a floating connection that is also an anti-lift connection. In examples, every beam-panel connection **214** of floating support grid system **200** may include a floating connection that is also an anti-lift connection. In examples, a beam-panel connection **214** may be provided to connect a lateral side (i.e. along the width or Y direction) of a support beam **204** to a lateral side edge **244***a* or **244***b* of a profile wire screen panel **202**. In examples, a beam-panel connection **214** between a support beam **204** and a profile wire screen panel **202** may be provided at or along a midsection and/or central portion of support beam **204**, a profile wire screen panel **202** or grid **222**, or a combination thereof.

[0086] As used herein, the term “midsection” when referring to the floating support grid system **200** as a whole, to a support beam **204**, to a profile wire screen panel **202**, and/or to a grid **222** or wires **224**, indicates an area that is not at a lateral (i.e. in the width or Y direction) peripheral edge of the structure. As used herein, the term “central portion” or “center” when referring to the floating support grid system **200** as a whole, to a support beam **204**, to a profile wire screen panel **202**, and/or to a grid **222** or wires **224**, indicates an area along line **246** that corresponds to the middle or approximately the middle of the width (i.e. dimension in the width or Y direction) of the referred

structure.
[0087] In examples, a longitudinal end connection **216** between a profile wire screen panel **202** and a portion **212** of a horizontal ammonia converter basket may be provided at or along a midsection and/or central portion of a lateral side edge end of the profile wire screen panel **202**. In examples, a longitudinal end connection **216** may be aligned with at least one beam-panel connection **214** for that same profile wire screen panel **202**. In examples, a floating support grid system **200** may be configured such that every beam-panel connection **214** between each profile wire screen panel **202** and respective support beam **204**, and every longitudinal end connection **216** between a profile wire screen panel **202** and a portion **212** of a horizontal ammonia converter basket are aligned.
[0088] In examples, the floating support grid system **200** can exhibit sufficient structural integrity to support the catalyst and operate as intended in a horizontal ammonia converter basket, while also being configured to allow for thermal expansion movements of one or more profile wire screen panels **202** and/or support beams **204**. In examples, this may result in reduced, or no stress imposed on the overall floating support grid system **200** in the longitudinal direction X, especially when compared to a catalyst bed in which the profile wire screen panels are welded joint to the beams as previously described with reference to FIGS. **1**A-**1**C. In examples, the floating support grid system **200** may also be configured to exhibit reduced or no stress imposed on the overall floating support grid system **200** in the width direction Y. In examples, the floating support grid system **200** may also be configured to exhibit reduced or no stress imposed on the overall floating support grid system **200** in the width direction Y and in the length or longitudinal direction X.
[0089] FIGS. **2**C and **2**D illustrate a diagram illustrating a side view of an example of a profile wire screen panel **202** supported by two support beams **204** (e.g., **204**′ and **204**″) as implemented in a floating support grid system **200** and of an example profile wire screen panel **202** as employed in a floating support grid system **200**.
[0090] In examples, as shown in FIG. **2**D, a profile wire screen panel **202** may include a grid **222** of one or more wires **224**. In examples, the wires **224** may include wedge wires, i.e. wires with a triangular and/or wedge-shaped cross-section. In examples, a profile wire screen panel **202** may include one or more support bars **226**. In examples, support bars **226** may be provided under one or more wires **224**, i.e. on the side of one or more wires **224** that is opposite the side of where a catalyst is held during use of the profile wire screen panel **202** when installed in a horizontal ammonia converter basket. In examples, a support bar **226** may be connected to one or more wires **224** and/or grid **222**. In examples, a support bar **226** may be arranged to have its longitudinal (i.e. X direction) side perpendicular to or substantially perpendicular to the lateral (i.e. Y direction) side of one or more wires **224**. In examples, the profile wire screen panel **202** may include one or more edge bars **228** (e.g., **228***a*, **228***b*, **228***c*, and **228***d*). In examples, an edge bar may extend along the full length of at least one side of grid **222**. In examples, an edge bar **228** (e.g., **228***a* and **228***b*) may be arranged to have its lateral (i.e. Y direction) side parallel or substantially parallel to one or more wires **224**. In examples, an edge bar **228** (e.g., **228***c* and **228***d*) may be arranged to have its longitudinal (i.e. X direction) side perpendicular or substantially perpendicular to one or more wires **224**. In examples, an edge bar **228** may be connected to the one or more support bars **226**. In examples, two edge bars **228** may be connected to one another. In examples, an edge bar **228** may be connected to one or more wires **224**. In examples, the connection between one or more support bars **226** and one or more wires **224**, grid **222**, and/or edge bar **228** may be a welded joint. In examples, the connection between one or more edge bars **228** and one or more wires **224**, grid **222**, and/or other edge bar **228** may be a welded joint.
[0091] In examples, a profile wire screen panel **202** as implemented in a floating support grid system **200** described herein may include one or more built-in plates **230**. In examples, as shown, a built-in plate **230** may be located at a peripheral area of a top surface **232** of grid **222** and/or one or more wires **224**. In examples, as shown, a built-in plate **230** may be located only at a peripheral area of grid **222** and/or one or more wires **224**. As used herein, a “peripheral area” refers to an outer

portion of a surface that surrounds a central portion of the surface. In examples, a built-in plate **230** may extend around the full perimeter of a top surface **232** of the grid **222** over one or more wires **224** (i.e. directly opposite side of wires **224** from where the one or more support bars **226** are located) of profile wire screen panel **202**. In examples, built-in plate **230** may extend only a portion of the perimeter. In examples, built-in plate **230** may include a single integral plate extending the full perimeter of top surface **232**. In examples, built-in plate **230** may include a set of two or more plates. In examples, a built-in plate **230** may extend over top surface **232** of grid **222** and/or one or more wires **224** from about 12 mm (or 0.5 inch) to about 75 mm (or 3 inch) as measured from a side edge of the grid **222** and/or one or more wires **224** toward the middle of grid **222** and/or one or more wires **224** for.

[0092] The profile wire screen panel **202** may include a metal or metal alloy. In examples, the one or more wires **224** of grid **222**, the one or more support bars **226**, the one or more edge bars **228**, and the built-in plate **230** may each independently include a metal or metal alloy material. In examples, the one or more wires **224** of grid **222**, the one or more support bars **226**, the one or more edge bars **228**, and the built-in plate **230** may all include the same material. In examples, the one or more wires **224** of grid **222**, the one or more support bars **226**, the one or more edge bars **228**, and the built-in plate **230** may each independently include the same or different material of any other portion of profile wire screen panel **202**. In examples, the one or more wires **224** of grid **222**, the one or more support bars **226**, the one or more edge bars **228**, and the built-in plate **230** may each independently include steel or other metal alloy.

[0093] In examples, as illustrated in FIG. **2**C, in floating support grid system **200**, a profile wire screen panel **202** may be connected to two support beams **204** by a floating connection that defines and/or includes an edge gap **234** between at least a portion of a lateral (i.e. along the width or Y direction) side edge **244***a* or **244***b* of profile wire screen panel **202** and a portion of the support beam **204** that is configured to support that lateral side edge **244***a* or **244***b* of profile wire screen panel **202**. In examples, edge gap **234** may extend between the profile wire screen panel **202** and the support beam **204** connected thereto along the full length of lateral side edge **244***a* or **244***b* of profile wire screen panel **202**. In examples, where a profile wire screen panel **202** is between two support beams **204** as shown in FIG. **2**C, an edge gap **234** may be present either between either lateral side edge **244***a* or **244***b* of profile wire screen panel **202** and respective support beam **204** (e.g., **204**′ and **204**″) that supports that side edge. In examples, where a profile wire screen panel **202** is between two support beams **204** as shown in FIG. **2**C, two edge gaps **234** (e.g. **234***a* and **234***b*) may be present, one between each lateral side edge **244***a* and **244***b* of profile wire screen panel **202** and respective support beam **204** (e.g., **204**′ and **204**″) that supports that side edge. In examples, where profile wire screen panel **202** includes an edge bar **228** as illustrated in FIGS. **2**C and **2**D, then edge gap **234** may be present between at least a portion of the edge plate **228** and support beam **204**. In examples, edge gap **234** may have a width A, i.e. shortest distance from a surface of support beam **204** to a surface at a lateral side edge **244***a* or **244***b* of profile wire screen panel **202** and/or of edge bar **228** if present. In examples, an edge gap **234** may range from about 3 mm (or ⅛ inch) to about 25 mm (or 1 inch). In examples, the edge gap **234** is provided between an outer surface **236** (e.g., **236***a* and **236***b*) of edge plate **228** and a vertical portion of beam **204**. In examples, the outer surface **236** of edge plate **228** is the surface that faces away from grid **222** of wires **224**.

[0094] In examples, a support beam **204** may include at least a portion forming an inverted T-shaped profile with a vertical web portion **238** (e.g., **238**′ and **238**″) extending upward from a top surface of a bottom horizontal base portion **240** (e.g., **240**′ and **240**″). In examples, edge gap **234** is located between a lateral side edge **244***a* or **244***b* of a profile wire screen panel **202** and a surface of vertical web portion **238** of a support beam **204**. In examples, where a profile wire screen panel **202** includes an edge bar **228**, the edge gap **234** may be located between outer surface **236** of an edge bar **228** and a surface of vertical web portion **238** of support beam **204**. In examples, as profile wire screen panel **202** moves laterally (i.e. in the longitudinal direction X) due to thermal expansion, edge gap **234** may accommodate the expansion thus avoiding structural impingement and thus stress to one or more components of the floating catalyst support system **200**.

[0095] In examples, as also shown in FIG. **2**C, the floating catalyst support system **200** may include one or more cover plates **242**. In examples, a cover plate **242** may be arranged and configured to prevent or impede debris or catalyst that is loaded over a profile wire screen panel **202** from reaching and/or entering edge gap **234**. In examples, a cover plate **242** may extend from a support beam **204**. In examples, a cover plate **242** may be welded, mechanically fastened by bolts or like structures, or both to a support beam **204**. In examples, a cover plate **242** may be provided on one or both sides of vertical web portion **238** of a support beam **204**. In examples, as shown in FIG. **2**C, each support beam **204**′ and **204**″ may include one or more cover plates **242** (e.g., **242**′ and **242**″) where a profile wire screen panel **202** is to be installed. In examples, a cover plate **242** may be arranged and configured to at least partially overlap the built-in plate **230** of a profile wire screen panel **202** when the profile wire screen panel **202** is connected to a support beam **204** in a floating catalyst support system **200**. In examples, a cover plate **242** may be arranged and configured to at least partially overlap the edge plate **228** of a profile wire screen panel **202** when the profile wire screen panel **202** is connected to a support beam **204** in a floating catalyst support system **200**. In examples, a cover plate **242** may be arranged and configured to overlap the edge plate **228** and at least partially overlap built-in plate **230** of a profile wire screen panel **202** when the profile wire screen panel **202** is connected to a support beam **204** in a floating catalyst support system **200**. In examples, the overlap between a cover plate **242** and built-in plate **230** may be of about 12 mm (or 0.5 inch) to 50 mm (or 2 inch).

[0096] An example beam-panel connection **214** that may be employed in the floating support grid system **200** is illustrated in FIGS. **3**A-**3**C. FIG. **3**A illustrates cross-section view of beam-panel connections **214***e* and **214***f* taken at A-A in FIG. **2**A. Beam-panel connections **214***e* and **214***f* are only used for illustrative purposes. All beam-panel connections **214** may be implemented as illustrated. FIGS. **3**B and **3**C respectively illustrate a cross-section side view proximate to beam-panel connection **214***e* of floating support grid system **200** taken from B-B in FIG. **2**B, and a plan view a beam-panel connections **214***e* and **214***f*.

[0097] In examples, as shown in FIG. **3**A, a first and second profile wire screen panel **202**, for example **202***c* and **202***d*, provided on both sides of a support beam **204**, e.g. **204***c*. As described earlier, each profile grid panel **202** may include a respective grid **222** (e.g. **222***c* and **222***d*) of wires **224** (e.g. **224***c* and **224***d*), respective one or more a support bars **226** (e.g. **226***c* and **226***d*), respective an edge bar **228** (e.g. **228***c* and **228***d*), and a respective built-in plate **230** (e.g., **230***c* and **230***d*). Support beam **204***c* may include an inverse T-shape profile including a vertical web portion **238** extending upward from a horizontal base portion **240**. As illustrated, the profile wire screen panels **202** may be supported at least at one edge portion by the support beam horizontal base portion **240** on either side of vertical web portion **238**. As also shown, an edge gap **234** (e.g. **234***c* and **234***d*) is formed between each profile wire screen panel **202** and the support beam **204**. In examples, ceramic rope **248** (e.g. **248***c* and **248***d*) may be provided in edge gap **234** between a profile wire screen panel **202** and a support beam **204**. In examples, ceramic rope **248** may include any suitable ceramic rope material. In examples, ceramic rope **248** can easily compressed. In examples, ceramic rope **248** may include fiberglass. In examples, ceramic rope **248** can help prevent catalyst and/or other debris from entering edge gap **234**.

[0098] In examples, as shown in FIG. **3**A and as previously described, floating support grid system **200** may include one or more cover plates **242** extending from one or more support beams **204**. In examples, shown in FIG. **3**A, respective cover plates **242***c* and **242***d* may be provided to extend over at least an edge portion of respective profile wire screen panels **202***c* and **202***d*. As previously described, a cover plate **242** may be configured to extend over at least a portion of built-in plate **230**. In examples, cover plate **242** overlaps built-in plate **230**. In examples, not binding connection is present between cover plate **242** and built-in plate **230**. In this manner, in examples, a profile wire screen panel **202** may undergo thermal expansion and/or contraction by moving, e.g., by sliding, between cover plate **242** and base portion **240** of beam **204**. In examples, the built-in plate **230** of a profile wire screen panel **202** has a width sufficiently sized to cover the maximum about of movement a profile wire screen panel **202** may experience. In examples, built-in plate **230** may protect grid **222** and/or wires **224** of profile wire screen **202** by preventing rubbing of cover plate **242** on grid **222** and/or wires **224** during movement caused by thermal expansion or restriction.

[0099] In examples, cover plate **242** may be provided to prevent catalyst loaded over the top surface of a profile wire screen **202** from reaching edge gap **234**. In examples, a floating support grid system **200** may be configured so that a gap between cover plate **242** and built-in plate **230** is less than a particle diameter of a catalyst to be loading on profile wire screen panel **202** during operation of the horizontal ammonia converter basket. In examples, a cover plate **242** may be positioned such that when profile wire screen panel **202** is connected to support beam **204** via the floating connection, a gap between cover plate **242** and built-in plate **230** prevents or impedes catalyst particles or debris from reaching edge gap **234**.

[0100] In examples, to ensure the gap between cover plate **242** and a built-in plate **230** is maintained substantially uniform and/or within a desired range, floating support grid system **200** may include one or more cleats **256**. In examples, a cleat **256** may be configured to extend from a side of vertical web portion **238** of a support beam **204** to a top surface of cover plate **242**, wherein the top surface of cover plate **242** is opposite the surface facing base portion **240** of support beam **204**. In examples, cleat **256** may include a wedge-shaped piece, however, it may include any desired shape. In examples, a cleat **256** may be made of the same or different material as support beam **204**. In examples, a cleat **256** may include a metal or metal alloy, for example steel. In examples, a support beam **204** may include one or more cleats **256** along width Y of floating support grid system **200**.

[0101] As illustrated in FIG. **3**A, a support beam **204**, e.g. **204***c*, may include one or more cleats **256**, e.g. **256***c* and **256***d*, one either or both side of vertical web portion **238**. In examples, one or more cleats **256** may be provided whenever a cover plate **242** is present. In examples, one or more cleats **256** may be provided at one or more locations along a lateral side of vertical web portion **238**. In examples, when more than one cleat **256** is present, they may be regularly spaced or irregularly spaced along the full lateral length of vertical web portion **238** and/or cover plate **242**.

[0102] In examples, a profile wire screen panel **202** may include an anti-lift system including an anti-lift bar **250**. In examples, anti-lift bar **250** may be provided at a bottom portion of a profile wire screen panel **202**. In examples, anti-lift bar **250** may be connected to one or more support bars **226** of a profile wire screen panel **202**. In examples, an anti-lift bar **250** may be connected to a bottom surface of one or more support bars **226** that is opposite the surface of the one or more support bars **226** facing and/or connected to grid **222** and/or wires **224**. In examples, anti-lift bar **250** may be connected directly to one or more support bars **226**. In examples, the connection between anti-lift bar **250** and one or more support bars **226** may be a welded joint. In examples, one or more connecting plates **252** may be placed between an anti-lift bar **250** and one or more support bars **226**. In examples, the anti-lift system may include a connecting plate **252**. In examples, a connecting plate **252** may be connected to two or more support bars **226** on a first side and to one or more anti-lift bars **250** on a second side opposite the first side. In examples, connecting plate **252** may have a profile that extends across an area that reaches two or more support bars **226**. In this manner connecting plate **252** may be more easily connected to two or more support bars **226**. In examples, at least one anti-lift bar **250** may be connected to connecting plate **252**. In examples, all connections between anti-lift bar **250** and either a support bar **226** or a connecting plate **252** may be fixed connections such as welded joints **254** and/or a mechanical fastened joint such as by screw and bolt or other like structure, or any combination thereof. Similarly, in examples. all connections between a support bar **226** and a connecting plate **252** may be fixed connections such as welded joints and/or a mechanical fastened joint such as by screw and bolt or other like structure, or any combination thereof. In examples, anti-lift bar **250** and connecting plate **252** may include the same or different materials used for one or more of the other components of profile wire screen panel **202**. In examples, anti-lift bar **250** and connecting plate **252** may each independently include a metal or metal alloy, for example steel.

[0103] In examples, as illustrated in FIG. **3**A, each of profile wire screen plates **202***c* and **202***d* may include an anti-lift bar **250** (e.g., **250***c* and **250***d*). In examples, each of anti-lift bars **250***c* and **250***d* may be connected to a respective connecting plate **252***c* and **252***d* via a respective fixed connection such as weld joins **254***c* and **254***d*. In examples, each connecting plate **252***c* and **252***d* may be connected to one or more respective support bars **226***c* and **226***d*. In examples, although not shown, connecting plates **252***c* and **252***d* are connected to respective one or more support bars **226***c* and **226***d* by a fixed connection such as a welded joint or bolt and screw type of fastening joint or a combination thereof.

[0104] In examples, as shown in FIG. **3**A, an anti-lift bar **250** may be configured to extend to reach below a bottom surface **257** of base portion **240** of a support beam **204**. In examples, anti-lift bar **250** may extend at least a distance a over bottom surface **257** of base portion **240**. In examples, distance a may range from about 25 mm (or 1 inch) to about 50 mm (or 2 inch). In examples, an anti-lift bar **250** may assist in preventing or hindering the lifting or vertical displacement of a profile wire screen panel **202**.

[0105] FIG. **3**B illustrates a cross-section side view to beam-panel connection **214***c* of floating support grid system **200** taken from B-B in FIG. **2**B. FIG. **3**B shows a vertical web portion **238** of support beam **204***c*, with two cleats **256** (e.g. **256***a* and **256***b*) extending diagonally from vertical web portion **238** to a cover plate **242**. As shown, cover plate **242** abuts but is not connected to built-in plate **230** of a profile wire screen panel **202**, e.g. **202***c*. Below built-in plate **230**, profile wire screen panel **202** includes grid **222** of wires **224**. Below grid **222**, profile wire screen panel **202** includes one or more support bars **226**. As illustrated, when profile wire screen panel **202** is connected to support beam **204**, at least an edge portion of profile wire screen panel **202** may rest of a base portion **240** of the inverted T-shaped support beam **204**. As illustrated, profile wire screen panel **202** may include a connection plate **252** fixed joined, e.g. welded, to one or more support bars **226**, for example at weld locations **258** (e.g. **258***a*, **258***b*, **258***c*). In examples, an anti-lift bar **250** may be fixed joined to connection plate **252**. In examples, the fixed joint **254** between anti-lift bar **250** and connection plate **252** may be one or more welded joints. As illustrated, anti-lift bar **250** may reach under base portion **240** of support beam **204**.

[0106] In examples, as shown in FIG. **3**B, floating support grid system **200** may include a beam-panel connection **214** with one or more stoppers **260** (e.g. **260***a* and **260***b*) to prevent movement of a profile wire screen panel **202** in a lateral or Y direction of the floating support grid system **200**. In examples, a stopper **260** may include any suitable material. In examples, a stopper **260** may include the same or different material as support beam **204**. In examples, stopper **260** may include a metal or metal alloy, for example steel. In examples, a stopper **260** may be joined to base portion **240** of a support beam **204** via one or more fixed connections **264** (e.g., **264***a* and **264***b*), e.g., by welding, by fastening mechanism such as bolt and screw or the like, or a combination thereof. In examples, a stopper **260** may be attached to a side surface **262** of base portion **240**. In examples, side surface **262** faces the same direction as the side of vertical web portion **238** where one or more cleats **256** are connected. In examples, as shown one or more stoppers **260** may be arranged to be on one or more lateral sides of an anti-lift system when the profile wire screen panel **202** is connected to support beam **204**. In examples, as shown one or more stoppers **260** may be arranged or provided to abut to anti-lift bar **250** and/or connection plate **252** of a profile wire screen panel **202** when the profile wire screen panel **202** is connected to support beam **204**.

[0107] FIG. **3**C illustrates a plan view of a bottom portion beam-panel connections **214***c* and **214***f* to further illustrate the arrangement of the anti-lift bars **250**. FIG. **3**C support beam **204***c* with vertical web portion **238** and a base portion **240**. For case of illustration, FIG. **3**C omits cleats **256**, cover plate **242**, and the profile wire screen panels **202***c* and **202***d* except for the anti-lift bars **250** and the connection plates **252**. As shown, attached to base portion **240** are one or more stoppers **260** (e.g. **260***a*, **260***b*, **260***c*, and **260***d*). FIG. **3**C also illustrates respective connection plates **252***c* and **252***d* that may be connected to one or more support bars **226** of profile wire screen panels **202***c* and **202***d*. FIG. **3**C further illustrates anti-lift bars **250***c* and **250***d* of profile wire screen panels **202***c* and **202***d* respectively. As shown, each anti-lift bar **250***c* and **250***d* may be connected to respective connection plate **252***c* and **252***d* by respective fixed connections **254***c* and **254***d*. Also, as shown, each anti-lift bar **250** may be configured to extent over a bottom surface of base portion **240** of support beam **204***c*.

[0108] FIG. **4** illustrates a longitudinal end connection **216** provided at longitudinal end portion **210** of the floating support grid system **200** where a profile wire screen panel **202** is directly connected to a portion **212** of a horizontal ammonia converter basket instead of a support beam **204**. In examples, longitudinal end connection **216** may also be configured to leave a panel end gap **266** between an edge portion **244***a* or **244***b* of profile screen wire panel **202** and portion **212**. In examples, panel edge gap **266** may extend between the profile wire screen panel **202** and portion **212** connected thereto along the full length of lateral side edge **244***a* or **244***b* of profile wire screen panel **202**. In examples, the panel end gap **266** may be the same or similar to the previously described edge gap **234**. In examples, the panel end gap **266** may range from about 6 mm (or 0.25 inch) to about 25 mm (or 1 inch).

[0109] In examples, longitudinal end connection **216** may include an end portion cover plate **268** that is similar to previously described cover plate **242** except that end portion cover plate **268** may be connected to portion **212** of the horizontal ammonia converter basket rather than a support beam **204**. In examples, the end portion cover plate **268** may be configured in the same or similar manner as previously described cover plate **242**. In examples, end portion cover plate **268** may be provided to prevent debris or catalyst particles provided over the top of profile wire screen panel **202** from entering panel end gap **266**. In examples, end portion cover plate **268** may be configured to overlap or abut without connection a top surface of built-in plate **230** of a profile wire screen panel **202**. In examples, the spacing between a bottom surface of end portion cover plate **268** facing built-in plate **230** and a top surface of built-in plate **230** facing the end portion cover plate **268** may be less than a diameter of a catalyst to be provided over profile wire screen panel **202**. In examples, the spacing may be configured to impede or prevent catalyst particles or other debris from reaching panel end gap **266**, as previously described with reference to cover plate **242**. In examples, to maintain the size of the spacing between end portion cover plate **268** and built-in plate **230** under a desired threshold, the floating support grid system **200** may include one or more end portion cleats **270** extending from end portion **212** to a top surface of end portion cover plate **268** in the same manner previously described for cleats **256**. In examples, end portion cleats **270** may be formed as brackets or may be designed as cleats **256**.

[0110] In examples, longitudinal end connection **216** of floating support grid system **200** may include a base ledge **272** protruding from portion **212** at a location below end portion cover plate **268**. In examples, base ledge **272** may be configured so that an edge portion of a profile screen wire panel **202** may rest thereon. In examples, when a profile wire screen panel **202** is connect to longitudinal end connection **216**, an end portion of profile wire screen panel **202** is configured to slide between a bottom surface of end portion cover plate **268** and a top surface of base ledge **272**.

[0111] In examples, an anti-lift bar **250** and optionally connection plate **252** of a profile screen wire panel **202** may be employed in the same manner previously described with reference to beam **204** by having anti-lift bar **250** extend over at least a portion of a bottom surface a base ledge **272** in the same manner as previously described with respect to the overlap of anti-lift bar **250** over a bottom surface of base portion **240** of a support beam **204**.

[0112] In examples, one or more end portion stoppers **274** may be provided to protrude from portion **212** of the horizontal ammonia converter basket and/or from base ledge **272** to prevent lateral movement in the Y direction of an installed profile wire screen panel **202**. In examples, end portion stoppers **274** may be configured and arranged in the same manner as previously described stoppers **260** except that end portion stoppers **274** may be connected to portion **212** instead of being connected to a base portion **240** of a support beam **204**.

[0113] In examples, the floating support grid system **200** may be configured to include and/or define one or more gaps or spacings to accommodate, at least in part, any lateral movement of one or more portions of floating support grid system **200** due to thermal expansion.

[0114] In examples, as shown in FIGS. **5**A-**5**C, the floating support grid system **200** may include a lateral ledge **282** configured to extend from a portion of horizontal ammonia converter basket along a lateral end portion of floating support grid system **200**. In examples, lateral ledge **282** may be provided as an integral part of horizontal ammonia converter basket. In examples, lateral ledge **282** can be fixed joint, e.g., by welding or a mechanical fastener such as bolt and screw or the like, or any combination thereof, to the horizontal ammonia converter basket. In examples, a lateral ledge **282** may be provided on either longitudinal side edge **244***c* and **244***d* of floating support grid system **200**. In examples, a lateral ledge **282** may be provided on both longitudinal side edges **244***c* and **244***d* of floating support grid system **200**. The description that follows with respect to lateral ledge **282** is equally applicable to either longitudinal side edge **244***c* and **244***d* of a floating support grid system **200**.

[0115] In examples, lateral ledge **282** may be configured such that when the floating support grid system **200** is installed at least a portion of lateral ledge **282** can overlap at least a longitudinal side edge **244***c* or **244***d* (i.e. an end portion along the X direction) of one or more profile wire screen panels **202**. In examples, the overlap between lateral ledge **282** and a lateral end of profile wire screen panel **202** may occur over at least a portion of a top surface of built-in plate **230** of the profile wire screen panel **202**. In examples, the overlap between lateral ledge **282** and a lateral end of profile wire screen panel **202** and/or built-in plate **230** may range from about 25 mm (or 1 inch) to about 50 mm (or 2 inch). In examples, the floating support grid system **200** may be configured such that a space or panel lateral side gap **286** is formed between a longitudinal (i.e. along the X direction) side edge **244***c* or **244***d* of a profile wire screen panel **202** and aa portion of a horizontal ammonia converter basket such as a wall or other structure **284** of a horizontal ammonia converter basket. In examples, panel lateral side gap **286** may extend between the profile wire screen panel **202** and a wall or other structure **284** of a horizontal ammonia converter basket connected thereto along the full length of longitudinal side edge **244***c* or **244***d* of profile wire screen panel **202**. In examples, as illustrated in FIG. **5**C, wall or structure **284** of a horizontal ammonia converter basket may include a bracket or like structure below lateral ledge **282**. This is just an example of structure **284** as other structures may be provided. For case of illustration, FIGS. **5**A and **5**B illustrate only longitudinal side edges **244***c* and **244***d* of two profile wire screen panels **202**. In examples, panel lateral side gap **286** may be configured to accommodate a thermal expansion movement in the Y direction of a profile wire screen panel **202**. In examples, panel lateral side gap **286** may be configured to accommodate movement in the Y direction of a profile wire screen panel **202** caused by the lateral thermal expansion of one or more support beams **204**. In examples, a ceramic rope, similar to previously described ceramic rope **248**, may be provided in panel lateral side gap **286**. In examples, as a support beam **204** expands in the lateral direction, it may cause movement of a connected profile wire screen panel **202** because of the one or more stoppers **260** that may be configured to maintain lateral alignment between profile wires screen panels **202** and support beams **204**. As such, a panel lateral side gap **286** may be helpful to accommodate such lateral movement of one or more profile wire screen panels **202**.

[0116] In examples, a floating support grid system **200** may include one or more second floating connection, referred to herein as beam floating lateral connection **276** between support beams **204** and the horizontal ammonia converter basket. In examples, a beam floating lateral connection **276** may include an anti-lift connection. In examples, as shown in FIG. **2**B, every support beam **204** in floating support grid system **200** may be connected to the horizontal ammonia converter basket. In examples, a support beam **204** may be connected to the horizontal ammonia converter basket at a lateral end portion **218** and/or **220** as illustrated earlier with reference to FIG. **2**A. In examples, a support beam **204** may be connected to the horizontal ammonia converter basket at both lateral end portions **218** and **220**. In examples, each support beam **204** of floating support grid system **200** may be connected to the horizontal ammonia converter basket at both respective lateral end portions **218** and **220** of floating support grid system **200**. In examples, a connection between a support beam **204** and the horizontal ammonia converter basket may include a beam floating lateral connection **276**.

[0117] FIGS. **5**A and **5**B provide perspective views of a beam floating lateral connection **276**. In examples, every connection between every support beam **204** and the horizontal ammonia converter basket may include a beam floating lateral connection **276**.

[0118] In examples, a beam floating lateral connection **276** may define and/or include one or more gaps between a support beam **204** and a horizontal ammonia converter basket when the support beam is connected to the horizontal ammonia converter basket. In examples, each of the one or more gaps defined and/or included in a beam floating lateral connection **276** may be configured to allow for or accommodate for a thermal expansion movement in the width or Y direction of support beam **204**, a profile wire screen **202**, or both.

[0119] In examples, a beam floating lateral connection **276** may define and/or include one or more beam connection spacing or gaps configured to allow for a thermal expansion movement of a support beam **204** in a Y direction of the floating support grid system **200**. In examples, beam floating lateral connection **276** may include one or more cover plates to prevent or hinder catalyst that is to be loaded on a profile wire screen panel **202** from entering the one or more beam connection spacing or gaps. In examples, the beam cover plates may be formed of any suitable material. In examples, beam cover plates may include the same or different material as support beam **204**. In examples, beam cover plates may include a metal or metal alloy, for example steel.

[0120] In examples, as shown in FIGS. **5**A and **5**B, a beam floating lateral connection **276** of a floating support grid system **200** may include one or more lateral ledge plates **288**. In FIG. **5**B, lateral ledge **282** is not shown for ease of illustration. In examples, a lateral ledge plate **288** may be fixed connected to a lateral ledge **282**. In examples, a lateral ledge plate **288** may be connected to a lateral ledge **282** by a welded joint, mechanical fastener such as by bolt and screw or like mechanism, or a combination thereof. In examples, lateral ledge plate **288** may be arranged to laterally extend from lateral ledge **282**. In examples, lateral ledge plate **288** may be arranged to extend from lateral ledge **282** towards the center of floating support grid system **200** and/or of a support beam **204**.

[0121] In examples, a beam floating lateral connection **276** of a floating support grid system **200** may include one or more abutting plates **290**. In examples, an abutting plate **290** may be fixed connected to a support beam **204**. In examples, the fixed connection may be a welded joint, mechanical fastener such as by bolt and screw or like mechanism, or a combination thereof. In examples, an abutting plate **290** may be provided at a lateral end portion of a support beam **204**. In examples, an abutting plate **290** may be provided at a lateral end portion of a cover plate **242** of a support beam **204**. In examples, an abutting plate **290** may be an integral portion of cover plate **242** and/or fixed connected to cover plate **242**. In examples, the fixed connection may be a welded joint, mechanical fastener such as by bolt and screw or like mechanism, or a combination thereof.

[0122] In examples, lateral ledge plate **288** and abutting plate **290** may include any suitable material. In examples, lateral ledge plate **288** and abutting plate **290** may include the same or different material. In examples, lateral ledge plate **288** and abutting plate **290** may include the same or different material as lateral ledge **282**, support beam **204**, or both. In examples, each of lateral ledge plate **288** and abutting plate **290** may independently include a metal or metal alloy, for example steel.

[0123] In examples, a lateral ledge plate **288** and an abutting plate **290** may be configured to be aligned. In examples, a lateral ledge plate **288** and an abutting plate **290** have a matching or the same or different cross-sectional shape and/or size. In examples, lateral ledge plate **288** and abutting plate **290** may be configured such that when a lateral end of a support beam **204** is connected to the horizontal ammonia converter basket, a beam lateral side gap **292** may be defined between and aligned with lateral ledge plate **288** and abutting plate **290**. In examples, beam lateral side gap **292** may be configured to accommodate lateral movement of a support beam **204**. In examples, the lateral movement may be due to thermal expansion. In examples, the size of beam lateral gap **292** may range from about 12 mm (or 0.5 inch) to about 25 mm (or 1 inch).

[0124] In examples, floating support grid system **200** may include a seal plate **294**. In examples, seal plate **294** may be fixed connected to lateral ledge plate **288**. In examples, seal plate **294** may be an integral portion of lateral ledge plate **288**. In examples, seal plate **294** may be fixed connected to abutting plate **290** and/or support beam **204**. In examples, seal plate **294** may be an integral portion of abutting plate **290**, cover plate **242**, and/or of support beam **204**. In examples, the fixed connection may be a welded joint, mechanical fastener such as by bolt and screw or like mechanism, or a combination thereof. In examples, seal plate **294** may be configured to overlap, i.e. extend over, at least a portion of ledge plate **288**, abutting plate **290**, or both. In examples, the overlap between seal plate **294** and either ledge plate **288** and/or abutting plate **290** may be at least about 20 mm.

[0125] In examples, seal plate **294** may be configured to extend over beam lateral side gap **292**. In examples, seal plate **294** may be configured to cover the top and side of a beam lateral side gap **292**. In examples, seal plate **294** may be configured to extend from over beam lateral side gap **292** to a top surface of cover plate **242** of support beam **204** and/or built-in plate **230** of a profile wire screen panel **202** connected to the support beam **204**. In examples, seal plate **294** may be configured to seal a beam lateral side gap **292** from catalyst or other solid debris. In examples, seal plate **294** may be configured to prevent or impede catalyst or other debris from entering lateral side gap **292**.

[0126] In examples, seal plate **294** may be configured to slide over lateral ledge plate **288** and/or abutting plate **290**. In examples, seal plate **294** may be configured to maintain its function as it slides over lateral ledge plate **288** and/or abutting plate **290**. In this manner, as support beam **204** lateral moves due to thermal expansion seal plate **294** is configured to slide to avoid hindering or significantly hindering the movement that is being accommodated by beam lateral side gap **292**.

[0127] In examples, a floating support grid system **200** may include at least one arrangement of a lateral ledge plate **288**, abutting plate **290**, and seal plate **294** as described at a beam floating lateral connection **276** of a support beam **204**. In examples, a floating support grid system **200** may include two arrangements of a lateral ledge plate **288**, abutting plate **290**, and seal plate **294** as described at a beam floating lateral connection **276** of a support beam **204**. In examples, each of a lateral ledge plate **288**, abutting plate **290**, and seal plate **294** may be provided on each side of a vertical web portion **238** of a lateral end portion of a support beam **204**. In these latter examples, where a seal plate **294** would be provided on both longitudinal sides on a lateral end portion of a vertical web portion **238**, the two seal plates **294** may be implemented as two separate seal plates or a single integral seal plate that extends from one side of vertical web portion **238** to the other side of vertical web portion **238** as for example illustrated in FIGS. **5**A and **5**B. In examples, a floating support grid system **200** may include at least one arrangement of a lateral ledge plate **288**, abutting plate **290**, and seal plate **294** as described at both lateral connections of a support beam **204**. In examples, a floating support grid system **200** may include two arrangements of a lateral ledge plate **288**, abutting plate **290**, and seal plate **294** as described at both lateral connections of a support beam **204**. In examples, each of a lateral ledge plate **288**, abutting plate **290**, and seal plate **294** may be provided on each side of a vertical web portion **238** of both lateral end portions of a support beam **204**.

[0128] In examples, as shown in FIGS. **5**A and **5**B, beam floating lateral connection **276** may define and/or include one or more support beam web lateral end gaps **296** and **298** between a lateral end portion **300** of support beam **204** and the horizontal ammonia converter basket and/or a wall or structure **284** thereof. In examples, support beam web lateral end gaps **296** and **298** may each independently range from about 25 mm (or 1 inch) to about 50 mm (or 2 inch). The wall or other structure **284** of the horizontal ammonia converter basket is not illustrated in FIG. **5**B for clarity. In examples, vertical web portion **238** may include a stepped profile at a lateral end portion **300** of a support beam **204**. In examples, the stepped profile of vertical web portion **238** at lateral end portion **300** of a support beam **204** may include an upper vertical surface **302** and a lower vertical surface **304**. In examples, upper and lower vertical surfaces **302** and **304** can face the horizontal ammonia converter basket and/or wall or structure thereof at a location where support beam **204** is laterally connected thereto. In examples, the stepped profile of vertical web portion **238** may be configured such that upper vertical surface **302** is recessed with respect to lower vertical surface **304**. In examples, the stepped profile of vertical web portion **238** may be configured such that when support beam **204** is connected to a horizontal ammonia converter basket, upper vertical surface **302** is at location that is spaced from and faces lateral ledge **282** while lower vertical surface **304** is located below lateral ledge **282**.

[0129] In examples, a first beam web lateral end gap **296** may be provided between a lateral end portion **300** of a support beam **204** and a lateral ledge **282**. In examples, a first beam web lateral end gap **296** may be between vertical web portion **238** and lateral ledge **282**. In examples, a first beam web lateral end gap **296** may be formed between the upper vertical surface **302** of vertical web portion **238** and a lateral ledge **282**. In examples, the floating support grid system **200** may be configured such that a beam floating lateral connection **276** as the support beam **204** expands, upper vertical surface **302** can move towards lateral ledge **282**. In examples, floating support grid system **200** may include one or more support beam lateral end covers **306** extending from upper vertical surface **302** of vertical web portion **238**. In examples, a support beam lateral end cover **306** may be configured to extend over and thus cover first beam web lateral end gap **296**. In examples, a support beam lateral end cover **306** may be configured to extend over at least a portion of lateral ledge **282**. In examples, the overlap of a support beam lateral end cover **306** and lateral ledge **282** may range from about 25 mm (or 1 inch) to about 50 mm (or 2 inch). In this manner, as the support beam **204** moves in a lateral direction because of thermal expansion support beam lateral end cover **306** can slide over lateral ledge **282**. In examples, a support beam lateral end cover **306** may be configured to prevent or impede catalyst and/or debris from entering first beam web lateral end gap **296**.

[0130] In examples, a support beam lateral end cover **306** may include the same or different material as support beam **204**, seal plate **294**, or both. In examples, support beam lateral end cover **306** may include a metal or metal alloy, for example steel. In examples, a support beam lateral end cover **306** may be welded to support beam **204** and/or vertical web portion **238**. In examples, support beam lateral end cover **306** may be separate structure from one or more seal plates **294**. In examples, support beam lateral end cover **306** may be fixed connected such by welding and/or by a mechanical fastener to one or more seal plates **294**. In examples, as shown in FIGS. **5**A-**5**C, a support beam lateral end cover **306** may be part of an integral structure that also includes one or more seal plates **294**.

[0131] In examples, one or more support beam lateral end cleats **308** may be provided extending from upper vertical surface **302** of vertical web portion **238** to a top surface of support beam lateral end cover **306**. In examples, a support beam lateral end cleats **308** may be configured in the same manner as one or more cleats **256** previously described. In examples, a support beam lateral end cleat **308** may be configured to ensure that a spacing between support beam lateral end cover **306** and a top surface of lateral ledge **282** overlapped by the support beam lateral end cover **306** is sufficiently small to prevent or impede catalyst or other debris from entering first beam web lateral end gap **296**.

[0132] In example, a second beam web lateral end gap **298** may be provided between a lower vertical face **304** of vertical web portion **238** of support beam **204** and a wall or other structure **284** (not shown in FIG. **5**B) of a of a horizontal ammonia converter basket. In examples, a second beam web lateral end gap **298** may be configured to be below lateral ledge **282** when support beam **204** is connected to the horizontal ammonia converter basket. In examples, the second support beam web lateral end gap **298** may be configured to accommodate for lateral movement of support beam **204** caused by thermal expansion. In examples, the floating support grid system **200** may be configured such that a beam floating lateral connection **276** as the support beam **204** expands, lower vertical face **304** under lateral ledge **282** may move towards a wall or other structure **284** of a horizontal ammonia converter basket.

[0133] As also shown in FIG. **5**B, in examples, the vertical web portion **238** may include a notch or passthrough **310** at one or more lateral end portions **300** thereof. In examples, notch or passthrough **310** may be formed by grinding, cutting or like manner, or alternatively may be provided by attaching a vertical tab extending vertically from base **240** in front of and spaced from vertical web portion **238**. In examples, notch or passthrough **310** may allow for the wrapping or placement of ceramic ropes **248** previously described. In examples, ceramic ropes **248** may be wrapped around the full lateral length of vertical web portion **238** at a position where edge gaps **234** are to be provided. In examples, the ceramic ropes **248** may extend from one lateral side of vertical web portion **238** to the other opposite lateral side of vertical web portion **238** by wrapping around lateral end portion **300** through notch or passthrough **310** as also illustrated in FIG. **5**C.

[0134] FIG. **5**C illustrates a side view of beam floating lateral connection **276** as described with reference to FIGS. **5**A and **5**B.

[0135] In examples, as shown in FIG. **5**C, a beam floating lateral connection **276** of floating support grid system **200** may be configured to provide a support beam base lateral end gap **312**. In examples, a support beam base lateral end gap **312** may range from about 12 mm (or 0.5 inch) to about 25 mm (or 1 inch). In examples, a support beam base lateral end gap **312** may be provided between a lateral end **314** of base portion **240** of support beam **204** and a wall or other structure **284** of a horizontal ammonia converter basket in which the floating support grid system **200** is to be installed. In examples, as shown in FIG. **5**C, support beam base lateral end gap **312** may be contiguous with panel lateral side gap **286**. In examples, as illustrated in FIG. **5**C, at beam floating lateral connection **276**, the lateral end **314** of base portion **240** may be configured to rest on a support **316**. In examples, support **316** may be fixed connected to and/or an extension of a horizontal ammonia converter basket. In examples, base portion **240** of support beam **204** may be configured such that when lateral end **314** rests on support **316**, a support beam base lateral end gap **312** may be defined between base portion **240** and a wall or other structure **284** of a horizontal ammonia converter basket. In examples, support beam base lateral end gap **312** may be configured to accommodate for lateral movement of support beam **204** due to thermal expansion.

[0136] In examples, as also illustrated in FIG. **5**C, a beam floating lateral connection **276** may include a beam anti-lift bar **278**. In examples, beam anti-lift bar **278** may extend from a bottom surface **257** of base portion **240** of support beam **204**. In examples, beam anti-lift bar **278** may be fixed connection to support beam **204** and/or base portion **240**. In examples, the fixed connection may be via a welded joint, a mechanical fastener such as bolt and screw or like structure, or a combination thereof. In examples, beam anti-lift bar **278** may include an L-shape profile. In examples, beam anti-lift bar **278** may be configured to reach below a mating structure **280** that is fixed connected to and/or an extension of a horizontal ammonia converter basket. By this arrangement, beam anti-lift bar **278** can assist in preventing or hindering the lifting of support beam **204** and/or of floating support grid system at floating lateral connection **276**. In examples, an anti-lift bar gap **318** may be defined between beam anti-lift bar **278** and mating structure **280** when

support beam **204** is connected to the horizontal ammonia converter basket. In examples, anti-lift bar gap **318** may be configured to accommodate for lateral movement of support beam **204** caused by thermal expansion. In examples, anti-lift bar gap **318** may range from about 25 mm (or 1 inch) to about 50 mm (or 2 inch). In examples, one or more beam stoppers **320** (e.g., **320***a*, **320***b*, etc. . . .) (shown in FIG. **5**C broken line for clarity, and also shown in FIG. **2**B) may be provided so as to be on one or both sides of anti-lift bar **278** when support beam **204** is connected to the horizontal ammonia converter basket. In examples, the one or more beam stoppers **320** may prevent movement in the longitudinal direction X of support beam **204**. In examples, beam stopper **320** may be designed in the same manner as previously described stoppers **260** and **274**. In examples, beam stoppers **320** may be fixed connected, i.e. welded or mechanically fastened to the horizontal ammonia converter basket or portion thereof, for example to support **316** as shown in FIG. **5**C.
[0137] It will be apparent to those skilled in the art that various modifications and variation can be made in the present invention without departing from the spirit or scope of the invention. Thus, it is intended that the present invention cover the modifications and variations of this invention provided they come within the scope of the appended claims and their equivalents.

Claims

1. A floating support grid system comprising: a plurality of support beams; a plurality of profile wire screen panels; a plurality of first connections each connecting one support beam of the plurality of support beams to one profile wire screen panel of the plurality of profile wire screen panels, wherein, at least one first connection of the plurality of first connections comprises a first floating connection configured to define a first gap between the one support beam of the plurality of support beams and the one profile wire screen panel of the plurality of wire screen panels the at least one first connection connects; and a plurality of second floating connections each configured to connect a support beam of the plurality of support beams to a horizontal ammonia converter basket.
2. The floating support grid system of claim 1, wherein the first gap is configured to accommodate a thermal expansion movement of a profile wire screen panel, a support beam, or both.
3. The floating support grid system of claim 1, wherein each first floating connection connects one side of the one profile wire screen panel to one side of the one support beam.
4. The floating support grid system of claim 1, wherein at least one profile wire screen panel of the plurality of profile wire screen panels is connected to a first support beam of the plurality of support beams at a first lateral side of the at least one profile wire screen panel and is connected to a second support beam of the plurality of support beams at a second lateral side, opposite the first lateral side, of the at least one profile wire screen panel.
5. The floating support grid system of claim 4, wherein one first floating connection is provided to connect the first support beam with the at least one profile wire screen panel at the first lateral side, and another first floating connection is provided to connect the second support beam with the at least one profile wire screen panel at the second lateral side.
6. The floating support grid system of claim 1, wherein every first connection comprises a first floating connection.
7. The floating support grid system of claim 1, wherein each profile wire screen panel of the plurality of wire screen panels comprises: a wire grid comprises one or more wires; at least one support bar provided below the wire grid; and a built-in plate over a peripheral area of the wire grid.
8. The floating support grid system of claim 7, the first floating connection further comprising a first cover plate configured to overlap at least a portion of the built-in plate over the peripheral area of the wire grid of the one profile wire screen panel connected to the one support beam, wherein the first cover plate extends from the one support beam toward the one profile wire screen panel

connected to the one support beam.

9. The floating support grid system of claim 8, further comprising one or more cleats extending from a vertical web portion of the one support beam connected to the one profile wire screen panel to a top surface of the first cover plate.

10. The floating support grid system of claim 1, wherein the first floating connection further comprises a first anti-lift system, wherein the first anti-lift system comprises an anti-lift bar fixed connected to a bottom surface the one profile wire screen panel, the anti-lift bar configured to extend below a bottom surface of a base portion of the one support beam connected to the one profile wire screen panel.

11. The floating support grid system of claim 10, further comprising one or more stoppers extending from the base portion of the one support beam connected to the one profile wire screen panel, the one or more stoppers arranged to be on one or more lateral sides of the first anti-lift system to prevent lateral movement of the one profile wire screen panel relative to the one support beam connected thereto.

12. The floating support grid system of claim 1, further comprising one or more third floating connections configured to connect at least one profile wire screen panel of the plurality of wire screen panels to the horizontal ammonia converter basket, wherein each third floating connection is configured to define a third gap between the at least one profile wire screen panel and a third or fourth portion of the horizontal ammonia converter basket it is configured to connect when the at least one profile wire screen panel is connected to the horizontal ammonia converter basket.

13. The floating support grid system of claim 12, wherein the third gap is configured to accommodate a thermal expansion movement of the at least one profile wire screen, a support beam of the plurality of support beams, or both.

14. The floating support grid system of claim 12, wherein every third floating connection comprises: a second cover plate arranged to extend over the third gap defined by third floating connection; and a second anti-lift system.

15. The floating support grid system of claim 1, wherein at least one second floating connection of the plurality of second floating connections is configured to define one or more second gaps between the support beam of the plurality of support beams and a second one or more portions of the horizontal ammonia converter basket when the at least one second floating connection connects a lateral end of the support beam of the plurality of support beams to the horizontal ammonia converter basket.

16. The floating support grid system of claim 15, wherein each of the one or more second gaps is configured to independently accommodate a thermal expansion movement of the support beam of the plurality of support beams, a profile wire screen panel of the plurality of profile wire screen panels, or both.

17. The floating support grid system of claim 15, wherein the support beam of the plurality of support beam comprises a first lateral end and a second lateral end, opposite the first lateral end, and wherein one second floating connection is arranged to connect the first lateral end to the horizontal ammonia converter basket and another second floating connection is arranged to connect the second lateral end to the horizontal ammonia converter basket.

18. The floating support grid system of claim 15, wherein every support beam of the plurality of support beams comprises a respective first lateral end with a respective second floating connection and a second lateral end, opposite the first lateral end, with a respective second floating connection, wherein every first lateral end and every second lateral end are connected to the horizontal ammonia converter basket.

19. The floating support grid system of claim 15, wherein the second floating connection is configured to define a panel lateral side gap between a profile wire screen panel connected to the support beam of the plurality of support beams and one of the second portions of the horizontal ammonia converter basket.

20. The floating support grid system of claim 19, wherein the panel lateral side gap is below a lateral ledge fixed connected to the horizontal ammonia converter basket.
21. The floating support grid system of claim 15, wherein the second floating connection further comprises one or more abutting plates each fixed connected to the support beam of the plurality of support beams and one or more lateral ledge plate extending form a lateral ledge fixed connected to the horizontal ammonia converter basket, wherein each abutting plate is aligned with a lateral ledge plate and configured to define a respective beam lateral side gap when the support beam of the plurality of support beams is connected to the horizontal ammonia converter basket.
22. The floating support grid system of claim 21, further comprising one or more second cover plates, each second cover plate configured to cover the respective beam lateral side gap.
23. The floating support grid system of claim 22, wherein the respective second cover plate is configured to overlap with at least a portion of the respective lateral ledge plate and to slide over the respective lateral ledge plate.
24. The floating support grid system of claim 23, wherein the second floating connection provided at a lateral end of a support beam comprises two abutting plates and two lateral ledge plates, wherein the two abutting plates and the two lateral ledge plates are arranged to define two separate beam lateral side gaps.
25. The floating support grid system of claim 24, further comprising an integral cover plate arranged to cover both separate beam lateral side gaps and extending over both lateral ledge plates.
26. The floating support grid system of claim 15, wherein the second floating connection is configured to define a first beam web lateral end gap between a vertical web portion of the support beam and a lateral ledge fixed connected to the horizontal ammonia converter basket.
27. The floating support grid system of claim 26, further comprising a third cover plate covering the first beam web lateral gap, wherein the third cover plate is configured to slide over the lateral ledge.
28. The floating support grid system of claim 15, wherein the second floating connection is configured to define a second beam web lateral gap between a vertical web portion of the support beam and one of the second portions of the horizontal ammonia converter basket, wherein the second beam web lateral gap is below a lateral ledge fixed connected to the horizontal ammonia converter basket.
29. The floating support grid system of claim 15, wherein the second floating connection is configured to define a support beam base lateral end gap between a base portion of the support beam and one of the second portions of the horizontal ammonia converter basket.
30. The floating support grid system of claim 15, wherein the second floating connection is configured to define an anti-lift bar gap between an anti-lift bar connected to a base portion of the support beam and a mating structure connected to the horizontal ammonia converter basket.
31. A floating support grid system comprising: a plurality of support beams, each support beam comprising a first lateral end and a second lateral end, opposite the first lateral end; a plurality of profile wire screen panels each connected via a respective first floating connection to at least one support beam of the plurality of support beams, each profile wire screen panel comprising: a first lateral side edge and a second lateral side edge; a first longitudinal side edge and a second longitudinal side edge; a grid comprising one or more wires; and a built-in plate over a peripheral area of the grid; one second floating connection configured to connect a first lateral side edge of a first profile wire screen panel to a horizontal ammonia converter basket; another second floating connection configured to connect a second lateral side edge of a second profile wire screen panel to the horizontal ammonia converter basket; and a third floating connection provided at each first lateral end of each support beam and at each second lateral end of each support beam, wherein the third floating connection is configured to connect each respective first lateral end and second lateral end to the horizontal ammonia converter basket, wherein, each first floating connection is configured to define a first gap between each profile wire screen panel and respective support beam

to which it is connected, wherein, each second floating connection is configured to define a second gap between the respective first lateral side edge or second lateral side edge and a first or second portion of the horizontal ammonia converter basket, wherein each third floating connection is configured to define respectively one or more third gaps between the first lateral end of each support beam and one or more third portions of the horizontal ammonia converter basket, between the second lateral end of each support beam and one of the third portions of the horizontal ammonia converter basket, between the first longitudinal side edge of each profile wire screen panel and one of the third portions of the horizontal ammonia converter basket, and between the second longitudinal side edge of each profile wire screen panel and one of the third portions of the horizontal ammonia converter basket, and wherein the first gap, the second gap, and the one or more third gaps are configured to accommodate a thermal expansion movement of one or more support beams, profile wire screen panels, or both.

32. A horizontal ammonia converter comprising: a horizontal ammonia converter basket; and a floating support grid system of claim 1.